Capacity-feasible CVRP clustering on an analog neutral-atom register (Track A)

Pasqal challenge — team DuQuantum — repository ‘qhack‘

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1. Model

Bottom line. The register decides which customers share a vehicle; each group is then routed exactly. One atom per (customer, vehicle) pair: 8 / 30 / 60 atoms on tiers S / M / L. On tier S the device returns the best grouping in 89–99 % of shots. On M it returns valid groupings but poor ones. On L it returns almost nothing valid. §5 has the numbers, §6 the reasons.

Why cluster-first. Encoding the full route (customer, vehicle, position) needs about N^2K atoms: 32 / 300 / 900 for S / M / L, against 80 available (DERIVATIONS Q1). Deciding only the grouping needs NK. The cost is a *decomposition gap*: the best grouping under our objective is not always the grouping of the shortest routes. It is 0 % on 5 of 11 instances and up to 21.7 % on the rest (§5, decomposition table).

Variables. N customers with demands d_i; K identical vehicles of capacity Q; d_{ij} the distance between customers i and j. The bit $x_{ik} = 1$ means customer i rides on vehicle k. A valid grouping has exactly one 1 in every customer’s row (“one-hot”).

Objective. We want groups whose members are close to each other, so we add up the distances between every pair of customers that share a vehicle:

$$f(x) = \sum_{k=1}^{K} \sum_{i<j} d_{ij}\, x_{ik}\, x_{jk}. \tag{1}$$

It costs us the decomposition gap above, and it is additive over groups, which is what lets an exact dynamic programme certify the model optimum on every tier (`cvrp/exact.py`).

Constraint 1 — every customer on exactly one vehicle. Enforced by a squared penalty with weight A:

$$A \sum_{i=1}^{N} \Big(1 - \sum_{k=1}^{K} x_{ik}\Big)^2. \tag{2}$$

Expanded, it puts the same linear term $-A$ on every atom and a coupling $+2A$ between the K atoms of one customer (a “clique”). It costs nothing in atoms; it costs energy scale: A must exceed every other term (below), and it is the largest coupling the register has to realise.

Constraint 2 — vehicle load $L_k = \sum_i d_i x_{ik}$ at most Q. We use unbalanced penalization (Montañez-Barrera et al. 2024), $\sum_k \left[-\lambda_1 (Q - L_k) + \lambda_2 (Q - L_k)^2\right]$, because it needs no slack atoms. On one-hot strings its linear part is the same constant for every valid grouping, so it drops out together with λ_1 (the “one-hot identity”, DERIVATIONS §4); what survives is a load-balance term

$$\lambda_2 \sum_{k=1}^{K} L_k^2 \;=\; \text{const} + 2\lambda_2 \sum_{k} \sum_{i<j} d_i d_j\, x_{ik} x_{jk}. \tag{3}$$

It costs us a guarantee: for $K = 2$ its minimum is capacity-feasible whenever a feasible split exists; for $K \geq 3$ it is not guaranteed. We therefore score feasible decodes only and report the feasibility rate (brief Pitfall 9.1).

The whole cost function is Eq. (1) + Eq. (2) + Eq. (3). Every coupling is positive ($d_{ij} \geq 0$, $2A > 0$, $2\lambda_2 d_i d_j > 0$) and every linear term is $-A$: exactly the pattern an analog register with repulsive C_6/r^6 interactions and one global detuning can produce. The pairs that interact are the $N \times K$ rook’s graph (same customer, or same vehicle); the capacity term adds no new pairs.

Weights. $A = \max_i \sum_{j \neq i} (d_{ij} + 2\lambda_2 d_i d_j) + 1$, which provably makes every minimiser one-hot (DERIVATIONS §4). $\lambda_2 = 0.15\, R/(2Q)$ with $R = \max_i \sum_j d_{ij}$, from a 40-seed brute-force scan. Brute force on the five S instances: the minimiser is feasible and optimal on 5/5; A survives one halving and fails at $A/4$; λ_2 is not binding on any S instance.

2. Variable accounting (brief Derive 4.1)

Encoding	count	S	M	L	fits 80?
assignment variables only	NK	8	30	60	yes
+ binary slack (textbook)	$NK + K\lceil \log_2(Q{+}1) \rceil$	14	45	80	yes (L = cap)
ours (`onehot-ub`)	NK	**8**	**30**	**60**	yes

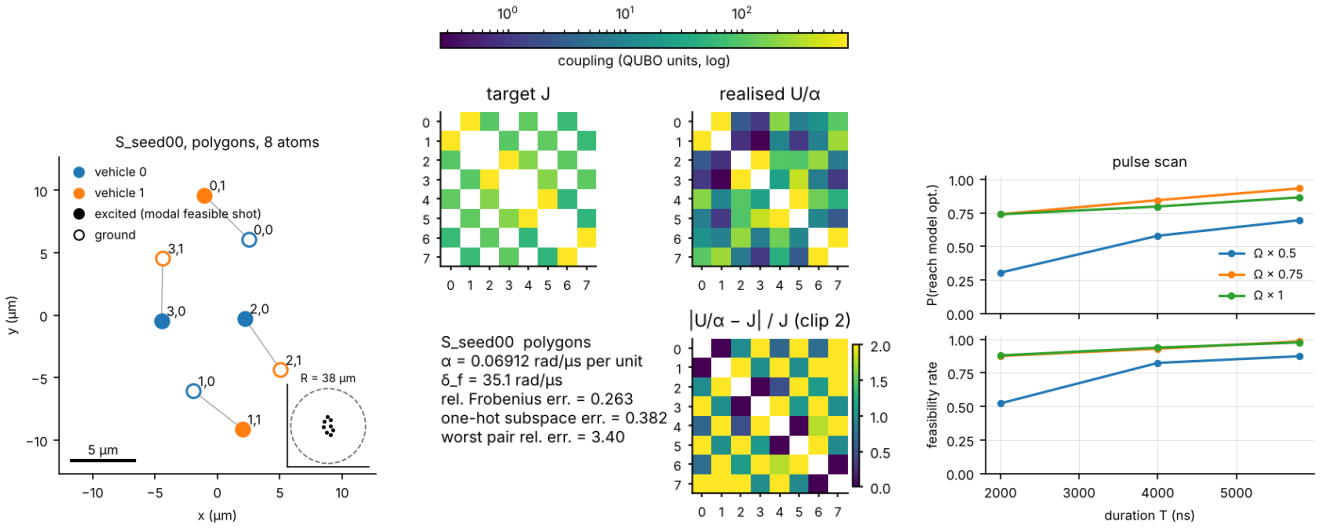


Figure 1: Left: S_seed00 register; each 5 µm dimer is one customer's exactly-one clique, filled atoms are the excited atoms of the modal feasible shot (one per dimer). Middle: target vs realised couplings; clique entries exact, cross-dimer entries not — the full-matrix coupling error (0.263) is dominated by pairs the fit does not need (worst pair 3.40), the one-hot-subspace error (0.382) by pairs it does. Right: pulse scan on three tier-S instances (in-house state-vector emulator, 500 shots, mean): longer is better at every Ω; 0.75 $\Omega_{\max}$ is on top at 4000 and 5800 ns; Ω/2 loses feasibility.

Tiers: S N=4, K=2, Q=7; M N=10, K=3, Q=20; L N=15, K=4, Q=30 (`scripts/variable_count.py`). We save 6 / 15 / 20 atoms over the textbook encoding and never discard a readout bit.

3. Embedding (DERIVATIONS §5)

Bottom line. A generic least-squares placement cannot realise the rook's graph. We fit the register only on the one-hot strings instead. On tier S ($K = 2$) that reproduces the model exactly; on M ($K = 3$) it reproduces the exactly-one constraint but not the ordering of groupings; on L ($K = 4$) the geometry of a square caps the energy scale below what the sweep can resolve.

Why generic placement fails. The K atoms of one customer must sit 5 µm apart (the clique). Any atom of another customer is then at nearly the same distance from all K of them, so the couplings that should be zero (different customer *and* different vehicle) come out as large as the wanted ones. Measured: relative coupling error 0.95, defined as $\|U/\alpha - J\|_F / \|J\|_F$ over all pairs (lower is better; `cvrp/embed.py`), and on S_seed04 the register's ground state was an infeasible string.

Ask for less. On a one-hot string each pair of customers has exactly one active coupling, so only the difference *same-vehicle minus cross-vehicle* has to match the model (DERIVATIONS 5.7b); the cross couplings only need to be constant. Each customer becomes a rigid 5 µm K-gon, which realises the clique exactly with $\alpha = C_6/(5\,\mu\text{m})^6/2A$; the centres and orientations are fitted (L-BFGS, 6 seeded starts) so that the device energies of all one-hot strings match $\alpha\, E_{\text{QUBO}}$, with single-flip defects forced above their parents and the final detuning δ_f as a fit parameter. Registers are committed in `embeddings/`; every run checks the realised register (is its own minimiser feasible and optimal? smallest defect gap). Tier S: 5/5 yes, coupling error 0.21–0.32.

A bound we found on tier M. At the end of the sweep ($\Omega = 0$) one atom of a K-gon excited costs $-\delta_f$; two excited cost $-2\delta_f + U_{\min}$, where $U_{\min}$ is the coupling of the *weakest* pair inside the polygon. One excitation per customer is the ground state only if

$$0 < \delta_f < U_{\min}. \tag{4}$$

For $K \leq 3$ every pair is the 5 µm side, $U_{\min} = 55.1$ rad/µs; the S fits land at $\delta_f = 35$–38 rad/µs (0.64–0.68 $U_{\min}$). On M the fit is ill-conditioned and ran to the device maximum, 125.66 rad/µs $= 2.28\,U_{\min}$, which favours *two* vehicles per customer; the embedder now enforces Eq. (4) (fallback $0.65\,U_{\min}$). For $K = 4$ a 5 µm square has 7.07 µm diagonals, and $1/r^6$ turns that $\sqrt{2}$ into a factor 8: $U_{\min} = 6.9$ rad/µs, so the L registers run at $\delta_f = 4.5$ rad/µs. Four points in a plane are never equidistant, so a planar one-hot clique for $K \geq 4$ is never uniform (DERIVATIONS §6.7).

4. Pulse (DERIVATIONS §6)

Chosen: 5800 ns (device maximum minus 200 ns), $\Omega = 0.75\,\Omega_{\max}$, fitted δ_f. Three-point Ω ramp (15 % up, 70 % plateau, 15 % down) and a linear detuning sweep during the plateau from $\delta_0 = -\min(\max(|\delta_f|, 2\Omega_{\max}), \delta_{\max})$ to δ_f. Starting far below resonance puts every atom in $|g\rangle$; ending at $\delta_f > 0$ with $\delta_f < U_{\min}$ makes one excitation per customer the target state. Every sequence is built with Pulser on `AnalogDevice` and validated against the device object.

Why these values. A scan on three S instances (Fig. 1 right; DERIVATIONS §6.6), scored by P(reach), best gap and feasibility, never mean cost: longer is better at every Ω (2000 → 4000 → 5800 ns); $0.75\,\Omega_{\max}$ beats full and half power at 4000 and 5800 ns; $\delta_f \times \{0.8, 1, 1.25\}$ gives 0.915 / 0.932 / 0.884, so the fitted value stays. S_seed02/03, not used in

the scan, reach the model optimum in 99 % / 96 % of shots with these settings. Emulators: pulser-simulation (QuTiP) for every S headline row; emu-sv cross-check on S_seed00 (0.96 vs 0.94); an in-house exact state-vector integrator built from Pulser's own samples for the scans (total-variation distance 0.05–0.07 from QuTiP at 500 shots, i.e. sampling noise); emu-mps for M and L with the bond dimension recorded.

5. Results (brief Table 3; full table: `results.csv`, 76 rows, `python -m cvrp.results`)

How to read a row. Raw shots, no repair. *feas.*: fraction of shots that decode to a capacity-feasible grouping. *model gap*: best feasible shot against the exact model optimum; *CVRP gap*: that shot routed exactly against C^* (proven on S, best known on M/L); both on feasible decodes only, "undefined" if there are none. *P(reach C^*)*: shots whose grouping routes to C^*, so it includes the decomposition loss. *P(gap ≤ 5 %)*: shots within 5 % of the model optimum; on every S instance and on M_seed00/01 this equals P(reach model optimum) because the next-best feasible grouping is 11–62 % worse, on M_seed02 it does not (six groupings within 5 %; DERIVATIONS Q-M3). *fidelity*: relative coupling error of the full matrix, lower is better.

instance	method	feas.	model gap	CVRP gap	P(reach C*)	P(gap≤5%)	fidelity	backend
S_seed00	ours	1.00	0.0 %	0.0 %	0.94	0.94	0.26	qutip
S_seed00	ours, tight A (ablation)	1.00	0.0 %	0.0 %	0.03	0.03	0.31	qutip
S_seed00	random assignments (one-hot)	0.63	0.0 %	0.0 %	0.12	0.12	0.26	random-onehot
S_seed00	qubo-solver quantum	1.00	0.0 %	0.0 %	0.78	0.78	0.12	qs-quantum
S_seed00	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed00	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.95	random
S_seed01	ours	0.95	0.0 %	0.0 %	0.95	0.95	0.21	qutip
S_seed02	ours	1.00	0.0 %	0.0 %	0.99	0.99	0.31	qutip
S_seed03	ours	1.00	0.0 %	15.6 %	0.01	0.96	0.29	qutip
S_seed04	ours	1.00	0.0 %	21.7 %	0.02	0.89	0.32	qutip
M_seed00	ours (emu-mps)	0.04	67.9 %	29.9 %	0.00	0.00	0.58	emu-mps χ=32
M_seed00	qubo-solver quantum	0.02	72.1 %	35.2 %	0.00	0.00	0.09	qs-quantum χ=32
M_seed01	ours (emu-mps)	0.39	37.9 %	9.2 %	0.00	0.00	0.08	emu-mps χ=32
M_seed01	qubo-solver quantum	0.37	47.5 %	11.2 %	0.00	0.00	0.08	qs-quantum χ=32
M_seed02	ours (emu-mps)	0.45	23.0 %	16.0 %	0.00	0.00	0.47	emu-mps χ=32
M_seed02	qubo-solver quantum	0.39	34.2 %	13.6 %	0.00	0.00	0.09	qs-quantum χ=32
L_seed00	ours (emu-mps)	0.01	140.0 %	79.7 %	0.00	0.00	0.65	emu-mps χ=32
L_seed01	ours (emu-mps)	0.00	182.9 %	75.1 %	0.00	0.00	0.65	emu-mps χ=32

(generated by `python scripts/report_tables.py` from `results.csv`; raw shots only — repair rows are in results.csv under their own label; shots: 500 per emulator row, 2000 for the random sampler, 1 for a classical solver; controls and ablations are shown for S_seed00 only — every other row of every method (random controls, tight-A ablation, tabu, pre-6.7 rows) is in results.csv)

Reference C^* (brief §9.3). CPLEX is not importable, so C^* is the best of qubo-solver's simulated annealing on our penalized QUBO and our constrained annealing on the CVRP, each 10 seeded restarts with a 60 s max solve time (settings in `instances/*.json`). On S the exhaustive dynamic programme is also run and agrees 5/5, so S references are proven; M/L are best known. The QUBO-side annealer reaches C^* on 5/5 S and 2/3 M instances but stays 23–34 % above it on all three L (599.9 / 554.6 / 521.7 vs 447.0 / 434.9 / 424.7), so C^* is never the QUBO solver alone.

instance	ρ	C* (best known)	model optimum routed exactly	decomposition gap
S_seed00	0.714	262.97	262.97	0.0 %
S_seed01	0.857	245.02	245.02	0.0 %
S_seed02	0.714	254.62	254.63	0.0 %
S_seed03	0.643	197.44	228.34	15.6 %
S_seed04	0.714	184.42	224.49	21.7 %
M_seed00	0.950	432.00	432.00	0.0 %
M_seed01	0.733	368.36	368.36	0.0 %
M_seed02	0.700	360.86	385.62	6.9 %

instance	ρ	C* (best known)	model optimum routed exactly	decomposition gap
L_seed00	0.650	446.97	525.71	17.6 %
L_seed01	0.708	434.91	470.67	8.2 %
L_seed02	0.600	424.66	480.17	13.1 %

Tier S — the device does the work. The model-optimal grouping is returned in 89–99 % of raw shots on all five instances, with 0.95–1.00 of shots feasible. The fair control — random assignment-valid strings through the same decoder — reaches it with probability 2/16 ≈ 0.12 on every instance; uniform random bitstrings 0.7–1.2 %; the generic embedder 3–13 %. qubo-solver's own pipeline (its embedding and drive, 5940 ns) reaches 0.78 / 0.78 / 0.18 / 0.79 / 0.00 and has a 27 % model gap on S_seed04; its tabu search finds every model optimum classically. Where P(reach C^*) is low with a 0 % model gap (S_seed03/04: 0.01 / 0.02) the loss is the decomposition gap (15.6 % / 21.7 %): the closeness objective's best grouping is not the shortest-route grouping. That is the price of §1, not of the device.

Ablation — the margin in A is physical. Replacing A by its tight provable bound (DERIVATIONS Q-A), same register construction and pulse, moves P(reach model optimum) from 0.94 to 0.03 on S_seed00, 0.95 to 0.52 on S_seed01 and 0.89 to 0.94 on S_seed04: a smaller A shrinks the defect gaps the sweep must resolve.

Tier M — valid groupings, poor ones. With δ_f bounded by Eq. (4), 98–100 % of shots are one-hot and 4.4 % / 38.8 % / 44.6 % are capacity-feasible on M_seed00/01/02; the best feasible shots are 67.9 / 37.9 / 23.0 % above the model optimum (29.9 / 9.2 / 16.0 % CVRP gap). A uniform random sampler decodes 0 of 2000 shots on every M instance. We enumerated all 59 049 one-hot groupings to locate the loss (DERIVATIONS Q-M3, `docs/checks/landscapes.md`): (i) the load-balance term leaves the model optimum at QUBO rank 1 on M_seed00/01, but on M_seed02 the QUBO minimum is a feasible grouping 10.2 % above it — exactly what qubo-solver's tabu search returns — so on M_seed02 about 10 of the 23 points are the λ_2 rule; (ii) under the realised register the model optimum sits at rank 10 474 / 12 966 / 4 192 of 59 049, which is the main loss; (iii) the emulated best shots beat the register's own best feasible string (82.0 / 101.3 / 93.9 %), so the sweep is not the bottleneck. qubo-solver's independent pipeline lands in the same regime (feasible 0.016 / 0.37 / 0.388, CVRP gap 35.2 / 11.2 / 13.6 %), while its tabu search on the same QUBO reaches 17.6 % / 2.8 % above C^* on M_seed01/02: the QUBO is solvable classically; the planar register limits it. Emulator check: M_seed00 needs a bond dimension of only 12, so $\chi = 32$ and $\chi = 64$ give the same untruncated run (feasible 0.044 vs 0.030, gap 67.9 vs 70.2 %: sampling noise); M_seed01 and L hit the $\chi = 32$ cap and are provisional (DERIVATIONS §6.8).

Tier L — almost nothing valid. L_seed00 / L_seed01 (60 atoms, emu-mps $\chi = 32$, 500 shots, $\delta_f = 4.475$ rad/µs): 1.4 % / 0.4 % of shots one-hot, 0.8 % / 0.4 % feasible, best feasible shot 140 % / 183 % above the model optimum (79.7 % / 75.1 % CVRP gap); both runs hit the $\chi = 32$ cap and are provisional (DERIVATIONS §6.8). The cause is Eq. (4): with $U_{\min} = 6.9$ rad/µs the whole one-hot landscape of 60 atoms is compressed into a few rad/µs (smallest defect gap 1.1 rad/µs against 19–23 on M), and a sweep of 30 rad/µs in 4 µs is far from adiabatic there (rate ≈ 7 rad/µs² against a gap-squared of ≈ 1.2). The model optimum sits at one-hot rank 16 534 / 1 063 / 14 862 of 1 048 576 on L_seed00/01/02 — on L_seed01 the register's own ground state is even capacity-feasible, and the sweep still does not reach it. L_seed02 is in `results.csv` as measured.

6. What did not work

- **Generic embedding of the one-hot graph.** Tried least-squares placement of all NK atoms; coupling error 0.95, an infeasible ground state on S_seed04, P(reach) 3–13 %. Cause: the clique forces the zero couplings to be as large as the wanted ones. Replaced by the one-hot-subspace fit (§3): P(reach) 0.89–0.99.
- **Unbounded δ_f on tier M.** The first M rows had $\delta_f = 125.66$ rad/µs $= 2.28\,U_{\min}$: 0 of 500 shots decodable on M_seed00/02. Cause: two excitations per triangle were favoured. Fixed by Eq. (4); the old rows stay in `results.csv` under `|pre-6.7`. Then the triangle version of the bound was applied to tier-L squares: the first L run ($\delta_f = 38.7$) was killed before reporting and the bound rewritten for the weakest pair.
- **The one-hot ordering for $K = 3$ and $K = 4$.** Opening the M triangles from 5 to 6–10 µm lowers the energy scale without fixing the ordering (optimum rank ≥ 1 279); a $K = 2$ bisection puts it at rank 192 594 / 12 217 (`docs/GATE2_OPTIONS.md`). We did not find a planar register that orders one-hot groupings by Eq. (1) for $K \geq 3$.
- **The λ_2 rule on M_seed02.** The load-balance weight moves the QUBO optimum 10.2 % above the model optimum on one of three M instances (0 % elsewhere). We kept the rule because changing weights after seeing results was ruled out; the effect is quantified in Q-M3.
- **Register reproducibility.** The polygon optimiser gave two different 30-atom registers from one seed when two runs overlapped (M_seed01, optimum rank 1 114 vs 12 966). Rule adopted: every reported row loads the committed register; one control row was re-measured.
- **Classical references.** CPLEX is not importable; the S agreement check is brute force vs two annealers. qubo-solver's default annealer is a single-start descent at $T_0 = 5$ and misses the S optima, so a stated-parameter annealer is used.
- **Weight scan.** A 12-point scan of (A, λ_2) around the rule showed no trend: the embedder's local minimum moves P(reach) more than the weights do (DERIVATIONS §4.1).
- **Tooling.** A `|` in a run tag lost one Windows record; a silent LaTeX failure left a stale PDF once. Both are caught by the build scripts now.